

Digital Signal Analysis and Processing (CT704) + Digital Signal Processing (EX753)

Concise PYQ 2066–2082

Notes: TU IOE PYQ. **CT704** (BEI/BCT, IV/I, 2066 Ma–2082 Bh, 27 papers) = normal font; **EX753** (BEX, IV/II, 2069 Bh–2081 Ch, 17 papers) = **typewriter**. Regular = **bold**, Back = normal. Marks in (); unique values only. Keys: (I)=Importance/Need (+)=Advantage (-)=Disadvantage +Fig=Figure +Eg=Example Vs=Compare/Differentiate. Ch. 1 and 9 are **EX753** only.

1. Intro to DSP, Sampling & Quantization (EX753)

- Basic elements of DSP +Fig; A/D–D/A conversion (I) (3+4)(4+2)(7)
- DSP (+) over analog; (-) of DSP; applications (2)(3)(6)
- Sampling theorem; Nyquist rate; aliasing; oversampling; anti-aliasing filter; sample-and-hold (2+5)(7)(1+3+3)
 - $x_a(t) = 3 \cos 100\pi t$ at $F_s = 75$ Hz / 200 Hz / 5000 Hz
 - multi-tone $x(t)$ sampled at 5000 / 6000 / 8000 / 10000 samples/s
- Quantization at 8 / 10 bits; discrete values + quantization error (2+5)(1+3+3)
- Spectral properties of time-domain sampling +Eg (4+2)

2. Discrete-Time Signals & Systems

- Even/odd DT signals +Eg; even/odd part; plot $x[-2n+3]$ +Fig (2)(3)(2+3)(4+5)
- Plot a composite sequence of $u[n], \delta[n], nu[n]$ (2)(3+2)
- Unit step +Fig; relation to unit impulse (1+2)
- Energy Vs power signal; check type; energy & power of $u[n]$ (2+2)(2+3)(3+4)(5)
- Periodicity + fundamental period (2+2)(3)(4)(5)
- System properties: linear / time-invariant / causal / memoryless / BIBO stable (3)(4)(5)(2+2+2)(3+3)(10)
- LTI system properties; recursive Vs non-recursive; moving-average recursive form (3+3)(5)(6)(8)
- Fourier series coefficients; discrete Fourier series of an impulse train; FT multiplication property; Dirichlet conditions (3)(4)(6)(4+3)
- Spectrum of CT Vs sampled signal +Fig (6)(8)
- Convolution summation — derive, significance (3)(1+3)(2+4)
- LTI convolution output (many h, x pairs: finite sequences, $u[n]$ gates, $a^n u[n]$, complex exponential inputs, tabulated / stem-plot data) (4)(5)(6)(7)(8)(3+7)(2+2+6)

3. Z-Transform

- ROC definition + properties + locate ROC (1)(2+4)(3+6)(4+6)(5)
- One-sided Z-transform (I); $Z\{na^n u[n]\}$; $Z\{\cos \omega n\}$ (5)(3+3)(3+8)
- Properties: convolution (state + prove), time-shift, time-delay, scaling (2+4)(3+3)(3+6)
- Inverse Z-transform by partial fraction (many $X(z)$, ROC given) (4)(5)(6)(1+5)(2+6)(2+2+3)
- Inverse Z-transform by long division (both ROCs) (6)(2+6)

4. Analysis of LTI in Frequency Domain

- Pole-zero plot + magnitude / phase response from a difference equation (many 2nd- and 3rd-order eqns) (6)(7)(8)(2+4)(3+7)(2+2+6)
- Same from given pole / zero locations ($0.45 \pm j1.06$, $0.58 \pm j2.06$, etc.) (8)(2+6)(2+8)(4+6)
- Frequency response from LCCDE; output for a sinusoidal input (6)(7)(10)
- FIR Vs IIR system (2)(4)
- LCCDE + system function; difference equation need (I) (2)(3)
- Zero-input / zero-state / total response, $H(z)$, poles (4)(10)
- Stability & causality via impulse response & ROC +Eg (2)(3)(4)(4+3)
- Linear-phase FIR condition; show $h = \{-1, 0, 1\}$ linear phase; derive freq. response for odd M (6)(2+4)

5. Discrete Filter Structures

- Direct Form I & II (many systems) (4)(5)(9)(2+2)
- Cascade form of 2nd-order sections; parallel form (4)(5)(6)(7)
- Lattice / lattice-ladder, FIR & IIR, + stability check (5)(6)(8)(9)(10)(6+1)(6+4)(7+2)

- 3-stage lattice $K_1, K_2, K_3 \rightarrow$ system function & FIR coefficients (5)(6)(7)
- all-pass filter via lattice (8)
- Limit cycle in recursive system +Eg; dead band; round-off effects (3)(4)(5)(1+3)(2+4)(3+4)
- Quantization (I); rounding Vs truncation; sign-magnitude / 1's / 2's complement; truncation-error bounds (6)(3+4)(1+1+2+1)

6. FIR Filter Design

- Window method + window characteristics; key points of windowing (4)(5)(4+4)(5+3)
- Gibbs phenomenon — how it arises with the rectangular window, how to minimize (3+3)(5)(6)(1+4)(2+5)
- Symmetric FIR LPF, $H_d(\omega) = e^{-j\omega\tau}$, length 7, $\omega_c = 1$ (Hanning / Blackman) (10)(2+7)(1+2+7)
- FIR LPF via suitable window from $\omega_p, \omega_s, \alpha_s$ (first 3 / first 6 coefficients) (6)(8)(9)(5+3)(2+6)
- Kaiser window design — ripple, β , length $M+1$ (many spec sets) (8)(10)(2+8)(8+4)(9+3)(2+2+2)
- Kaiser (+) over fixed windows; FIR Vs IIR choice (2)(3)(4)
- Optimum filter / Remez exchange — math expression, derivation, flowchart, (I) over windowing (4)(5)(6)(7)(9)(1+6)(3+10)
- Linear phase FIR from kHz specs (2 kHz / 5 kHz / 42 dB / 20 kHz) (10)

7. IIR Filter Design

- Bilinear-transformation Butterworth LPF — many spec sets, given as $|H|$ bounds, as dB deviations, or in Hz (9)(10)(11)(12)(15)(18)(11+4)
 - most common: $W_p = 0.25\pi$, $W_s = 0.55\pi$, $\delta_p = 0.11$, $\delta_s = 0.21$, $f_s = 0.5$ Hz, then LP $\rightarrow$ HP at $W'_p = 0.45\pi$
 - find N , ω_c , poles s_k , $H(s)$, $H(z)$ (part-wise version)
- Bilinear-transformation Chebyshev LPF; analog Chebyshev then convert (9)(10)(11)
- Impulse-invariance Butterworth; pre-warping (I); limitations (12)(13)(15)(12+3)
- Bilinear Vs impulse invariance (+); why transform a CT design (2+10)(3+6)(3+12)
- Butterworth Vs Chebyshev approximation (1)
- Spectral (digital-domain) transformation LP $\rightarrow$ HP; features / parameters; (I) (2)(4)(5)

8. Discrete Fourier Transform (DFT / FFT)

- DFT need (I); DFT Vs DTFT; complexity DFT Vs FFT; FFT (+) (1)(2)(4)(5)(2+5)(4+2)
- Circular convolution property (state + prove); multiplication property; zero padding; Parseval (1)(2)(6)(2+2+4)
- DFT duality $\rightarrow$ compute IDFT with the DFT formula (8)
- FFT (DIT / DIF) N -point DFT + butterfly diagram + spectrum (many sequences; several papers print 7 entries for an 8-point DFT) (5)(6)(7)(8)(1+6)(2+6)(3+7)(2+8)
- IDFT via FFT from a given $X(k)$ (2+6)
- Circular convolution — direct or via DFT (many pairs) (4)(5)(6)(7)(1+5)(2+5)(2+6)(3+7)
- Linear convolution through circular convolution with zero padding (5)(1+5)(2+6)(6+2)
- $x_3 = \text{IDFT}(X_1 \cdot X_2)$, 4-pt & 5-pt (5)(6)(7)(1+5)(2+4)(2+6)

9. DSP Processors (EX753)

- Bit-serial arithmetic +Fig; distributed arithmetic (5)(6)(3+3)
- Pipelined implementation of DSP processors (6)
- DSP processor chips — where used, general Vs special purpose, features (6)(2+4)